Multivariable Optimization

ECON 441: Introduction to Mathematical Economics
Lecture 17

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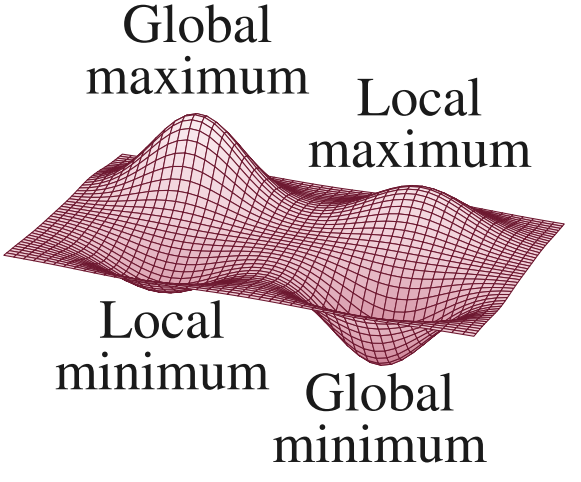
More than One Choice Variable

$$z = f(x, y)$$

What pair of values for x and y maximizes or minimizes this function?

We continue to restrict ourselves to continuous functions with continuous first derivatives.

More than One Choice Variable



First-Order Conditions

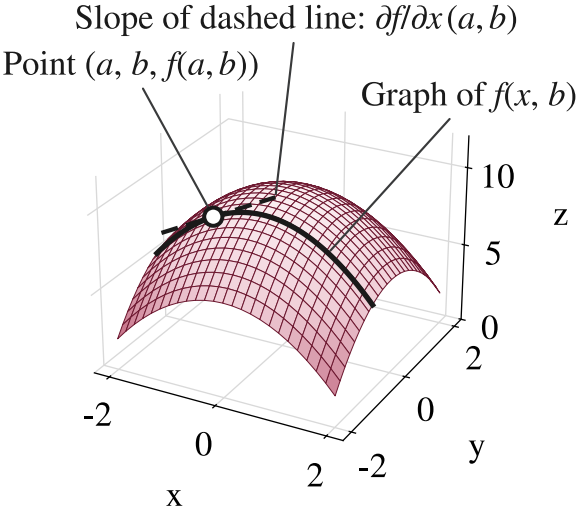
For the function

$$z = f(x, y)$$

The first-order (necessary) condition:

$$f_x = f_y = 0$$

Partial Derivative

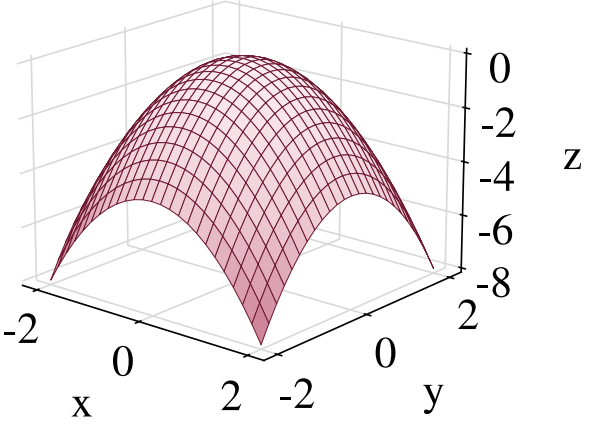


Example

What are the critical points of

$$f(x, y) = -(x^2 + y^2)$$

$$f(x, y) = -(x^2 + y^2)$$



Example

What are the critical points of

$$\pi(K, L) = f(K, L) - rK - wL$$

OLS

$$\min_{\alpha,\beta} \quad f(\alpha,\beta) = \sum_{i=1}^{n} (Y_i - \alpha - \beta X_i)^2$$

First-order conditions:

$$\frac{\partial f}{\partial \alpha} = -2 \sum_{i=1}^{n} (Y_i - \alpha - \beta X_i) = 0$$

$$\frac{\partial f}{\partial \beta} = -2 \sum_{i=1}^{n} (Y_i - \alpha - \beta X_i) X_i = 0$$

Second-Order Partial Derivatives

For the function

$$z = f(x, y)$$

$$f_{xx} \equiv \frac{\partial}{\partial x} f_x \quad \text{or} \quad \frac{\partial^2 z}{\partial x^2} \equiv \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)$$

$$f_{yy} \equiv \frac{\partial}{\partial y} f_y \quad \text{or} \quad \frac{\partial^2 z}{\partial y^2} \equiv \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right)$$

Second-Order Partial Derivatives

We also have cross (or mixed) second-order partial derivatives:

$$f_{xy} \equiv \frac{\partial^2 z}{\partial x \partial y} \equiv \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)$$

$$f_{yx} \equiv \frac{\partial^2 z}{\partial y \partial x} \equiv \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)$$

We always have $f_{xy} = f_{yx}$ as long as f_{xy} and f_{yx} are both continuous.

Example

Find the four second-order partial derivatives of

$$z = x^3 + 5xy - y^2$$

Second-Order Conditions

At a critical point, the second-order (sufficient) conditions are:

- Maximum: $f_{xx} < 0$, $f_{yy} < 0$, and $f_{xx}f_{yy} > (f_{xy})^2$
- Minimum: $f_{xx} > 0$, $f_{yy} > 0$, and $f_{xx}f_{yy} > (f_{xy})^2$

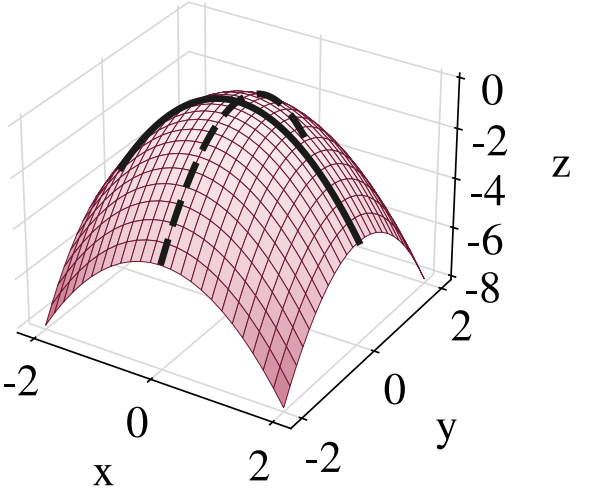
If $f_{xx}f_{yy} < (f_{xy})^2$, the point is a **saddle point**: neither a maximum nor a minimum.

Example

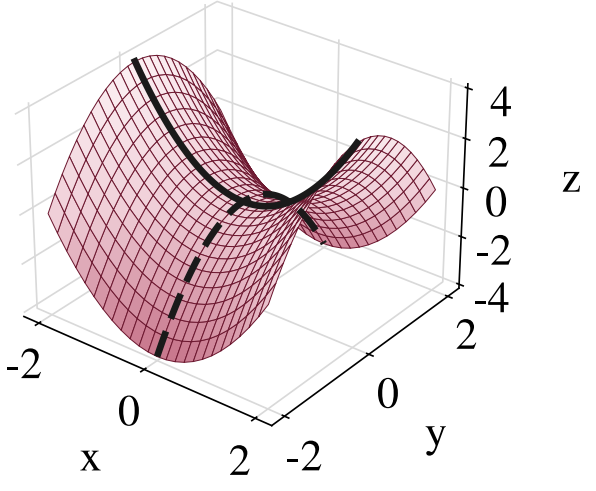
Find the critical point of each function and determine whether it is a maximum, a minimum, or neither.

1. $f(x, y) = -(x^2 + y^2)$
2. $f(x, y) = x^2 - y^2$
3. $f(x, y) = x^2 + y^2 + 5xy$

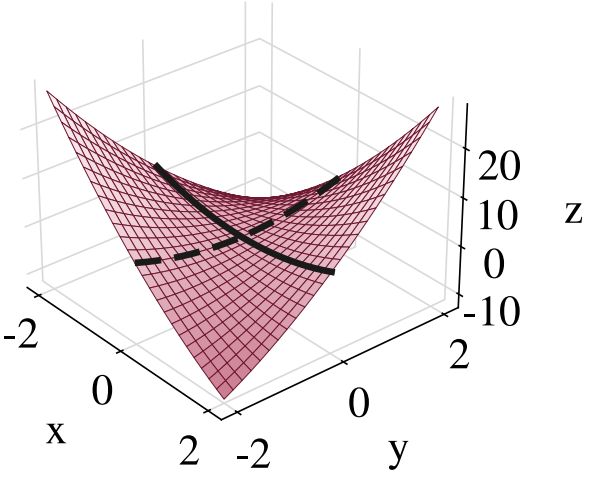
$$f(x, y) = -(x^2 + y^2)$$



$f(x, y) = x^2 - y^2$



$$f(x, y) = x^2 + y^2 + 5xy$$



Hessian Matrix

For the function

$$y = f(x_1, x_2, \dots, x_n)$$

The gradient vector ∇f and Hessian matrix H are

$$\nabla f = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix} \qquad H = \begin{bmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ f_{21} & f_{22} & \cdots & f_{2n} \\ \vdots & \vdots & & \vdots \\ f_{n1} & f_{n2} & \cdots & f_{nn} \end{bmatrix}$$

More than Two Choice Variables

- First-order necessary, for a maximum or a minimum: $f_1 = f_2 = \cdots = f_n = 0$, that is, $\nabla f = 0$
- Second-order sufficient for a maximum: $|H_1| < 0$, $|H_2| > 0$, $|H_3| < 0$, …
- Second-order sufficient for a minimum: $|H_1|, |H_2|, \ldots, |H_n| > 0$

Here $|H_k|$ is the determinant of the upper-left $k \times k$ block of H. The second-order conditions apply only after the first-order condition is satisfied.

Homework Questions

- Exercise 11.2: 1, 2, 3, 4, 5

Textbook reference: 11.1, 11.2